

Arjunaa Seshadri

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Professional Summary

LLM/AI Engineer with production experience shipping multi-agent systems, RAG pipelines, and fine-tuned models using LoRA/PEFT. Depth in agentic orchestration (LangGraph, CrewAI), fine-tuning and quantization (Qwen, Unsloth), and delivery deployment on AWS. Adept at Backend frameworks and DBMS workloads with scalability.

Education

Stony Brook University | M.S. in Data Science

Stony Brook, NY | August 2026 – Expected 2028

Jain University | B.Tech, Computer Science (AI & ML)

Bengaluru, India | August 2021 – May 2025

Experience

TechAhead | AI/ML Developer (*promoted from AI Engineer Intern*)

Noida, India | July 2025 – January 2026

- Architected multi-agent event scheduling workflows in CrewAI, LangGraph, and Redis, cutting scheduling turnaround 40% against the manual baseline.
- Shipped three Dockerized FastAPI microservices to AWS EC2 — video annotation, embedding generation with Qdrant storage, and the core chatbot — sustaining 120 req/s at 180 ms p95 under Locust load testing.
- Cut invalid Text-to-SQL generations 15–20% with LLM-as-a-Judge, via guardrail-based query validation, with Redis caching removing redundant model API calls.
- Rebuilt healthcare and fitness advisory flows as tool-calling agents on Amazon Strands and the OpenAI Agents SDK.

SVAAS Inframax Solutions | Data Science Intern

Noida, India | January 2025 – May 2025

- Built an AI scheduling assistant on Mistral 7B wired to the Google Calendar API, removing manual appointment entries for the operations team.
- Automated invoice ingestion with a PyTesseract OCR and LLM summarization pipeline, cutting per-invoice handling from 4 minutes to under 30 seconds.
- Deployed a YOLOv8 PPE detection system in PyTorch flagging real-time safety non-compliance at the sites.
- Shipped a production, two-portal operations platform used by the active employees for attendance, payroll, and inventory tracking.

CSIR - National Aerospace Laboratories | Project Vision Intern

Bengaluru, India | June 2024 - July 2024

- Built a YOLO + VisDrone computer vision POC performing spatial 3D object mapping from drone-mounted footage for UAV surveillance applications.

Projects

Blood Report Analyser — Agentic Clinical Report Pipeline | *LangGraph, GPT-4o, FastAPI*

- Engineered a 7-node LangGraph pipeline with parallel per-panel extraction (Send API) and a self-critique/retry loop, cutting end-to-end report processing latency by around 50%.
- Built a GPT-4o ReAct agent that cross-correlates markers across panels to surface multi-marker clinical syndromes invisible to single-panel review.
- Designed a longitudinal trend engine projecting threshold-crossing dates from patient history, plus a human-in-the-loop interrupt for unstructured data.

DataQuery — Text-to-SQL Analytics Platform | *Qwen 8B, Unsloth, Next.js, FastAPI*

- Fine-tuned and quantized Qwen 8B on Spider (5,000+ NL-SQL pairs), lifting execution accuracy on multi-join queries 10% over base.
- Architected a multi-tenant Next.js and FastAPI stack with row-level RBAC preventing cross-tenant data leakage.
- Built dynamic dashboard generation (Chart.js) from natural-language prompts, letting non-technical users self-serve data exploration without writing SQL.

SVAAS Employee Tracking System — Production HR & Ops Platform | *React/TS, FastAPI, PostgreSQL*

- Shipped a live two-portal platform used by the current employees covering a payroll engine reconciling GPS-verified attendance, leave policy, and salary advances, backed by JWT token rotation and four-tier RBAC.
- Built a server-validated inventory ledger blocking over-issuance at write time, plus a role-scoped OpenAI ops chatbot.

Technical Skills

Languages & Frameworks: Python, SQL, TypeScript, PyTorch, TensorFlow, Scikit-learn, Pandas, FastAPI, React

LLM & Agentic: LangGraph, LangChain, CrewAI, OpenAI Agents SDK, Hugging Face, Unsloth, vLLM, Qdrant

Infrastructure: AWS (EC2, CloudWatch), Docker, Nginx, Redis, PostgreSQL, MongoDB, Supabase, Git, Vercel

Certifications: IBM RAG & Agentic AI; Microsoft Certified: Azure AI Fundamentals